

Product Requirements Document

Clothing Material E-commerce Platform & Companion Admin App

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Prepared for: Intern Development Team

Field	Details
Client / Brand Name	TBD — placeholder brand name to be used until final naming is confirmed
Document Owner	Dominal Technology
Audience	Intern developers (Web + Flutter Android)
Platforms in scope	E-commerce website (customer-facing) + Android Admin App (Flutter)
Backend	Supabase (Postgres, Auth, Storage)

1. Project Overview

This document defines the requirements for a clothing/cloth-material e-commerce website and its companion Android admin application. The platform sells clothing material (fabric-based products) organized by gender, category, style, and material type. The brand name is not finalized — interns may use any placeholder brand name during development. Dummy product images (one per product type) should be used for now.

The system consists of two applications sharing one Supabase backend:

- **Customer Website** — browsing, filtering, cart, checkout, and profile management.
- **Admin App (Flutter, Android)** — full CRUD management of categories, products, images, and orders.

2. Design & UI Guidelines

2.1 Visual Style

- Clean, minimal interface. Background: white. Primary text: black.
- One accent / dark theme color used sparingly — roughly **20% of the UI surface** (buttons, active states, highlights), with white/black making up the remaining 80%.
- No emoji anywhere in the UI. Use custom **SVG icons** for all iconography (cart, profile, filter, menu, etc.).
- Typography and spacing should stay simple and generous — avoid visual clutter.

2.2 Navigation

- A **hamburger menu** is compulsory, placed in the top-left or top-right corner of the header (final side at dev discretion, but consistent across all pages).
- A persistent **bottom tab bar** with 3–4 primary options (e.g. Home, Categories, Cart, Profile). Exact icons/labels are up to the intern team, using SVG icons only.

2.3 CSS Architecture

- Only **one global CSS file** is allowed for shared styling.
- All colors, spacing, and font sizes must be defined as **CSS variables** (e.g. `--color-bg`, `--color-text`, `--color-accent`, `--spacing-sm`) — no hardcoded hex values or magic numbers inside components.

- Component-level styles should reference these variables so the entire theme can be re-skinned by editing one place.

3. User Profile Module

Each user account must support a profile containing:

- **Name**
- **Address**
- **Phone number**

3.1 Profile Completion Logic (Required Business Rule)

When a user clicks "**Add to Cart**" or attempts to place a **Buy Order**, the system must check whether the profile is complete. If **any single field** — name, address, or phone number — is missing, the user must be redirected to the profile page (or shown an in-app notification/toast) prompting them to "*Create or complete your profile*" before the action can proceed.

4. Information Architecture — Product Catalog

The catalog structure below reflects the initial scope (Men's category), based on the attached handwritten notes. The same structure pattern should be extensible to other genders/sections in future.

4.1 Men's Category Structure

Style	Sub-types / Patterns
Formal	Whites, Plain, Strips (Stripes), Checks, Prints
Casual	Checks, Whites, Solid, Prints

4.2 Material Types (applies across categories)

- Cotton
- Linen
- Khadi
- Silk
- Linen Cotton

4.3 Sample Reference Pricing (from notes — for dummy data)

Material	Reference Price
Khadi	Rs. 600
Silk	Rs. 350

Note: these two prices were the only ones specified in the source notes; interns should generate reasonable dummy prices for the remaining materials/products.

4.4 Filtering & Sorting

- **Price sort:** Low to High and High to Low.

- Additional filters to be added as needed (material type, style, pattern) — "and much more in the buying section" per stakeholder notes; interns should propose a filter set (material, pattern, price range) for review.

5. Product Listing, Product Detail & Cart

- **Product Listing page** — grid of products with image, name, material, price, and quick add-to-cart.
- **Product Detail page** — full product info, material, pattern/style, price, quantity selector, add-to-cart / buy now.
- **Cart page** — line items, quantity edit, remove item, subtotal, proceed to checkout.
- **Add to Cart / Buy Now** actions must trigger the profile-completion check described in Section 3.1.

6. Database & Migrations (Supabase)

All schema changes must be tracked as SQL migration files, stored in a dedicated **/migrations** folder in the repository. This folder acts as the running history of every SQL statement executed against the Supabase project to date.

- One file per migration, sequentially numbered and timestamped, e.g. `0001_create_profiles_table.sql`, `0002_create_products_table.sql`.
- Each migration file should contain only the SQL run for that change (idempotent where possible).
- The migrations folder must always reflect the true, current state of the Supabase schema — no undocumented manual changes.
- Suggested core tables: `profiles`, `categories`, `products`, `product_images`, `cart_items`, `orders`, `order_items`.

7. Admin Panel — Flutter Android App

The admin panel is a separate **Flutter Android application**, connected to the same Supabase backend, used internally to manage the storefront.

7.1 Core Capabilities

- Full **CRUD** (Create, Read, Update, Delete) operations for products.
- Manage **categories** — ability to add more categories beyond the initial Men's Formal/Casual set.
- Manage **product images** — upload, replace, delete multiple images per product.
- Manage material types, patterns/styles, and pricing.
- View and manage incoming orders and cart activity (recommended, for completeness).
- Same visual language as the website: clean white background, black text, ~20% accent/dark color theme, SVG icons instead of emojis.

7.2 Notes

The admin app should be a fully manageable control center — "everything" configurable from it, per stakeholder direction — so the storefront never requires direct database edits for day-to-day catalog changes.

8. Dummy Content & Branding

- Brand name is a placeholder — any name can be used for development purposes.
- Upload at least **one dummy image per product type/pattern** (e.g. one for Formal Whites, one for Formal Checks, one for Casual Solid, etc.) to populate the catalog for testing.

9. Requirements Summary

Area	Requirement
UI Theme	White background, black text, ~20% accent/dark color
Icons	SVG only — no emojis
Navigation	Hamburger menu (top corner) + bottom tab bar (3–4 items)
Profile	Name, Address, Phone Number
Profile Gate	Block Add-to-Cart / Buy if any profile field is missing; show completion prompt
Catalog	Men → Formal / Casual, each with pattern sub-types
Materials	Cotton, Linen, Khadi, Silk, Linen Cotton
Filters	Price low→high / high→low, plus additional filters
CSS	Single global CSS file, variables only, no hardcoded values
Database	Supabase, with full SQL migration history in /migrations
Admin App	Flutter Android app, full CRUD for products/categories/images
Dummy Data	Placeholder brand name, 1 image per product type

This PRD is derived from stakeholder handwritten notes and a verbal walkthrough. Items marked as recommendations or defaults should be confirmed with the client before final implementation.